

Random Forest

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Index

Random forest
↳ Decision Tree

- What is Decision Tree?
- Tree splitting ✓
- What is Random Forest? ✓
- Bagging ✓
- Features importance ✓

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Inverted tree



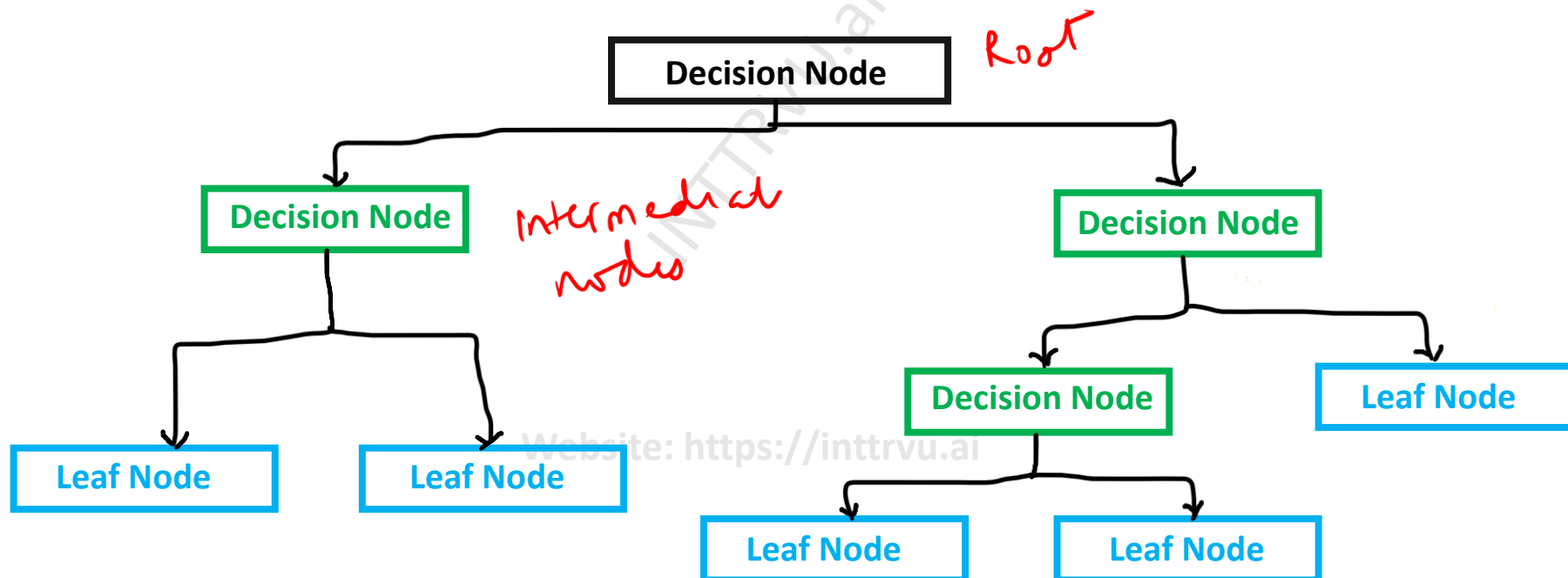
What is Decision Tree?

is a tree which helps you in making decisions

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What is Decision Tree?

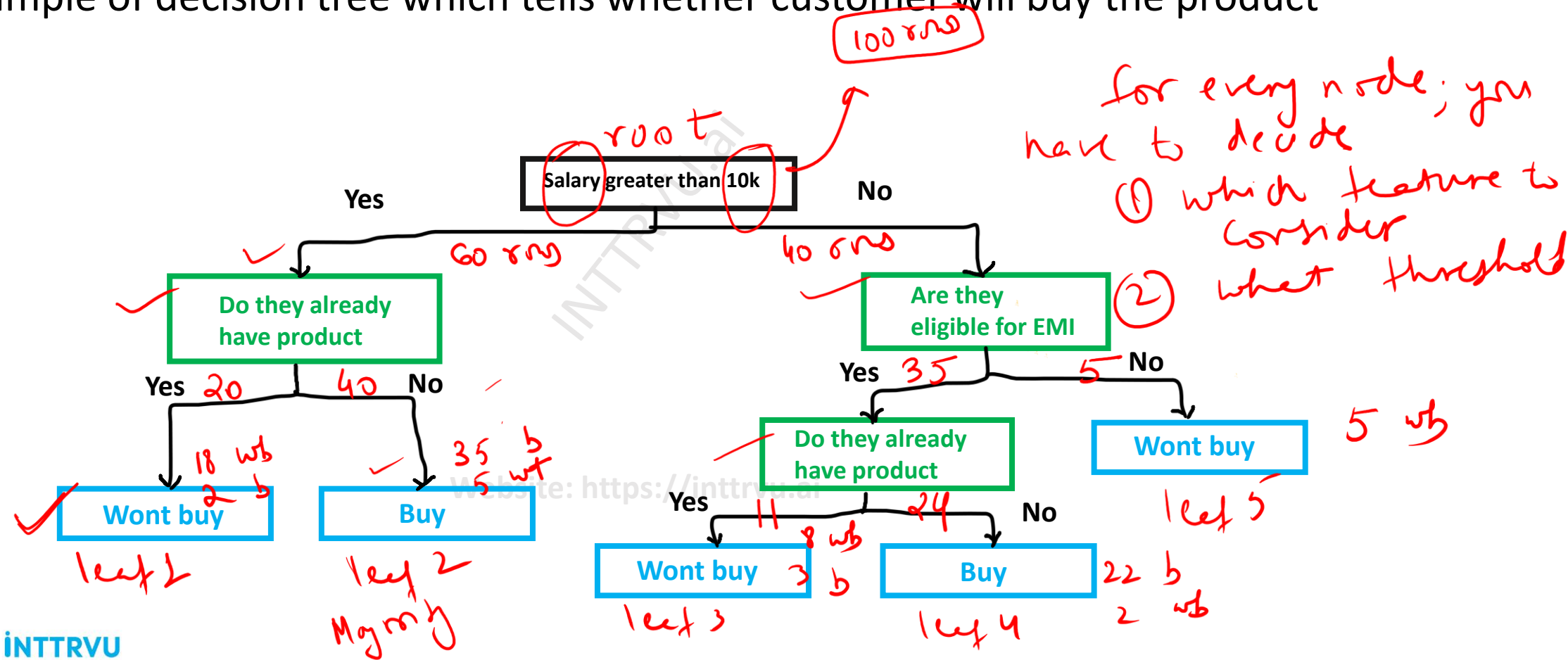
- It is a tree like structure which represents the way in which data is segmented based on rules



Q: whether a customer will buy the product or not
Features: Salary > 10k, Do they already have product, EMI eligibility

What is Decision Tree?

- Example of decision tree which tells whether customer will buy the product



iphone

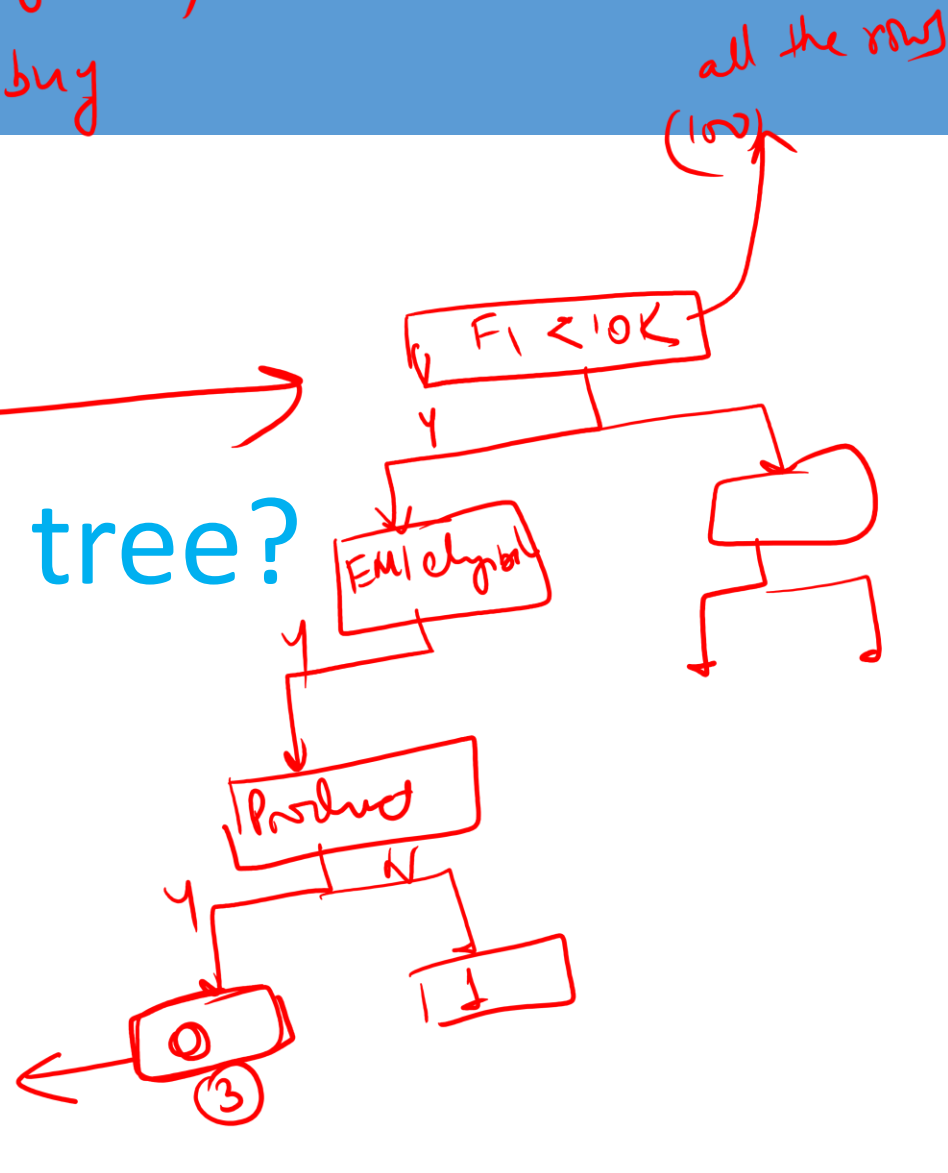
F ₁	F ₂	F ₃	Y
			0
			1
			1
			0
			0
			0

F₁ = salary
F₂ = customer has the product
F₃ = EMI eligibility
0 ⇒ won't buy
1 ⇒ buy

Entire data

To make a tree

How to split the tree?



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Tree splitting – Gini impurity

① Which feature to use for splitting
② What threshold to use for splitting

- ① • Gini impurity is one of widely used metric to decide best splitter for the decision tree (for classification tasks)
- ✓ • Gini impurity indicates how pure the split would be
- ✓ • Gini impurity is zero if after the split all underlying samples belong to single class
- ④ • Splitter with lowest gini impurity is chosen at every decision node in tree

✓ Gini impurity = $1 - \sum_{i=1}^k p_i^2$

✓ p_i is the proportion of i^{th} class label

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Gini Impurity tells you about the purity of a node

Root
Node

$$\begin{array}{l} 64 \rightarrow \frac{1}{2} \\ 36 \rightarrow 0 \end{array}$$

How do I judge the purity of this root node?

$$G = 1 - \sum_{i=1}^N p_i^2$$

N = number of classes available in a node

$$N=2 \Rightarrow G = 1 - (p_1^2 + p_2^2) = 1 - (0.36^2 + 0.64^2)$$

$$p_1 = \frac{36}{64+36} = 0.36, \quad p_2 = \frac{64}{36+64} = 0.64$$

Higher the value, higher will be the impurity.

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leaf

20	→	0
0	→	1

$$G = 1 - \sum p_i^2 = 0$$

$$G = 1 - [p_1^2] = 0$$

$$p_1 = \frac{20}{20+0} = 1$$

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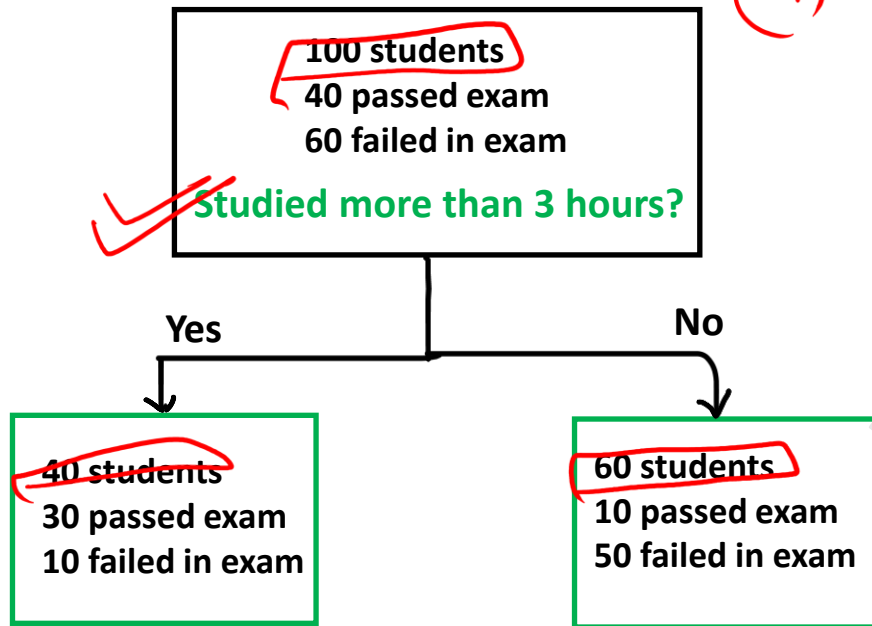
Gini impurity

$F_1 \mid F_2$

$F_1 \Rightarrow$ Study at home (hours)
 $F_2 \Rightarrow$ Attendance in class

$F_1 > Th_1$

$(F_1, Th_1) \Rightarrow$ one of the options



At Node A

Probability of passing = $30/40 = 0.75$

Probability of failing = $10/40 = 0.25$

Gini impurity = $1 - (0.75)^2 - (0.25)^2 = 0.38$

At Node B

Probability of passing = $10/60 = 0.17$

Probability of failing = $50/60 = 0.83$

Gini impurity = $1 - (0.17)^2 - (0.83)^2 = 0.28$

✓ Node A $\frac{40}{100}$
 G_A

✓ Node B $\frac{60}{100}$
 G_B

Gini impurity

G_A, G_B

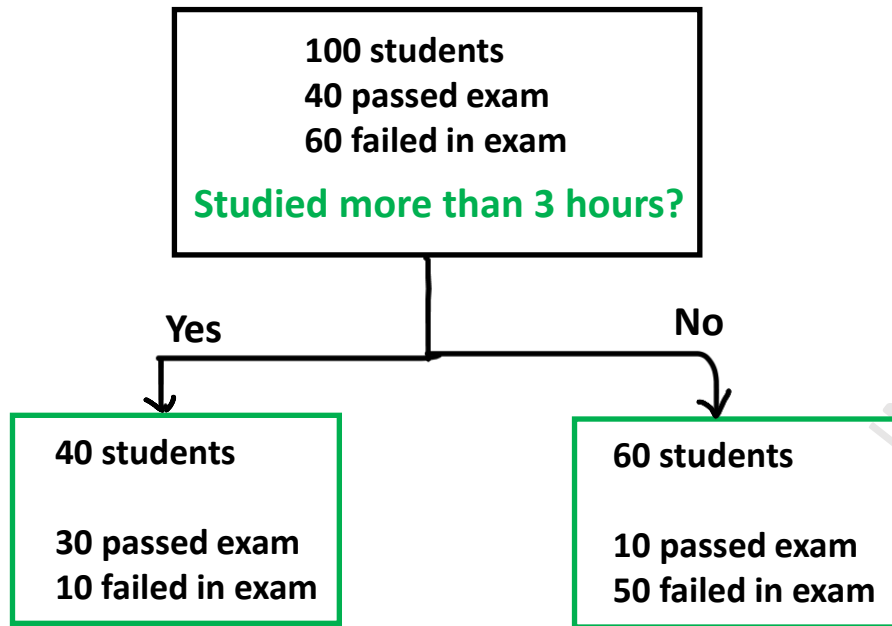
Gini Impurity score for (F_1, Th_1)

$(F_1, Th_1) \rightarrow 0.32$

$(F_1, Th_2) \rightarrow \sim$

$(F_1, Th_3) \rightarrow \sim$

$(F_2, Th_4) \rightarrow \sim$



Node A

Node B

$$\text{Gini impurity node A} = 1 - (0.75)^2 - (0.25)^2 = 0.38$$

$$\text{Gini impurity node B} = 1 - (0.17)^2 - (0.83)^2 = 0.28$$

Weighted Gini Impurity of split:

$$= (40/100) * 0.38 + (60/100) * 0.28$$

$$= 0.32$$

If there are multiple features/thresholds on which split can be performed, then the feature with lowest weighted Gini impurity will be used

Tree splitting – Information gain

- Entropy is a measure of uncertainty or randomness in a dataset.
- In decision trees, entropy is used to quantify the impurity of a node.
- A node with low entropy means the samples in that node are predominantly of a single class (high purity), while high entropy indicates a mixture of classes (low purity).

c is number of classes ← **c**

$$\text{Entropy} = \sum_{i=1}^c -p_i * \log_2(p_i)$$

Lower the purer

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Entropy = $-\sum_{i=1}^C p_i \log p_i \Rightarrow C$ is total number of classes in the node.
 p_i = probability of class 'i'

root 60 $\rightarrow$ 0
40 $\rightarrow$ 1

$$\Rightarrow \text{Entropy} = -\sum_{i=1}^2 p_i \log p_i = -\left[\frac{60}{60+40} \log \frac{60}{60+40} + \frac{40}{60+40} \log \frac{40}{60+40} \right]$$

$$= -[0.6 \log 0.6 + 0.4 \log 0.4]$$

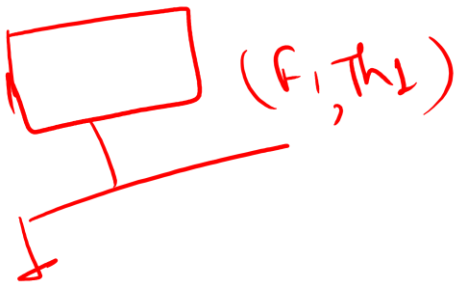
leaf 20 $\rightarrow$ 0
0 $\rightarrow$ 1 $\Rightarrow$ Entropy = ?? $-\frac{20}{20+0} \log \frac{20}{20+0} = -\log 1 = 0$

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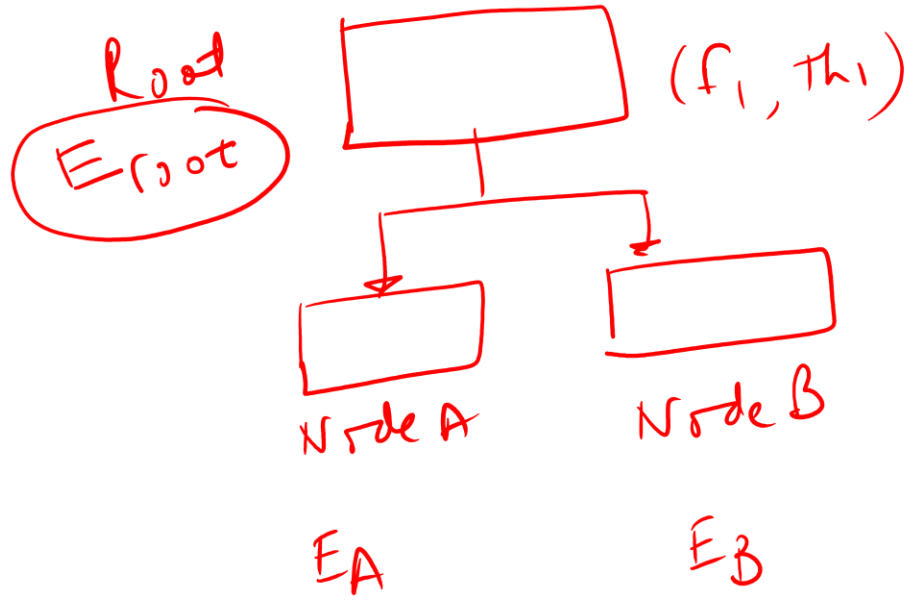
Tree splitting – Information gain

- The goal of entropy-based splitting is to maximize the information gain at each node.
- Information gain measures the reduction in entropy achieved by splitting a node.
- The splitting criterion selects the feature that maximizes information gain.

Information Gain (IG) = Entropy before splitting - Weighted average of entropies after splitting



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E_{split} = weighted average of E_A and E_B

Information gain - $E_{root} - E_{split} = \Delta E$

✓ Choose that splitter which will lead to max ΔE

$(f_1, Th_1) \rightarrow \Delta E_1$

$(f_1, Th_2) \rightarrow \Delta E_2$

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40	→ 0
50	→ 1
10	→ 2

$$G = 1 - [0.4^2 + 0.5^2 + 0.1^2]$$

$$E = -[0.4 \lg 0.4 + 0.5 \lg 0.5 + 0.1 \lg 0.1]$$

1

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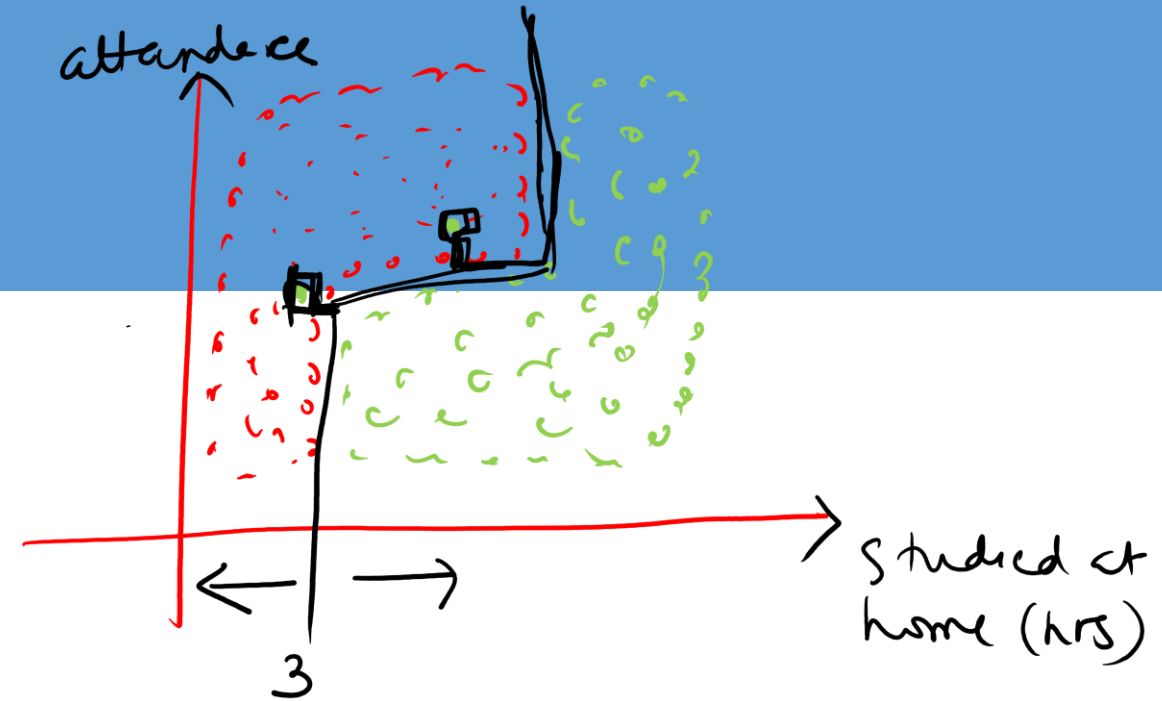
Decision Tree

Advantages:

- Easy to interpret and explain to stakeholders
- Data scaling /normalization is not must
- Can learn linear as well as non-linear decision rules

Dis-advantages:

- Slight change in data can bring big change in the tree structure. Data scaling /normalization is not must
- **Single decision tree is prone to overfitting**



Forest is a group of trees
Every decision tree is a model.

Random forest is a model which is a combination of multiple models.

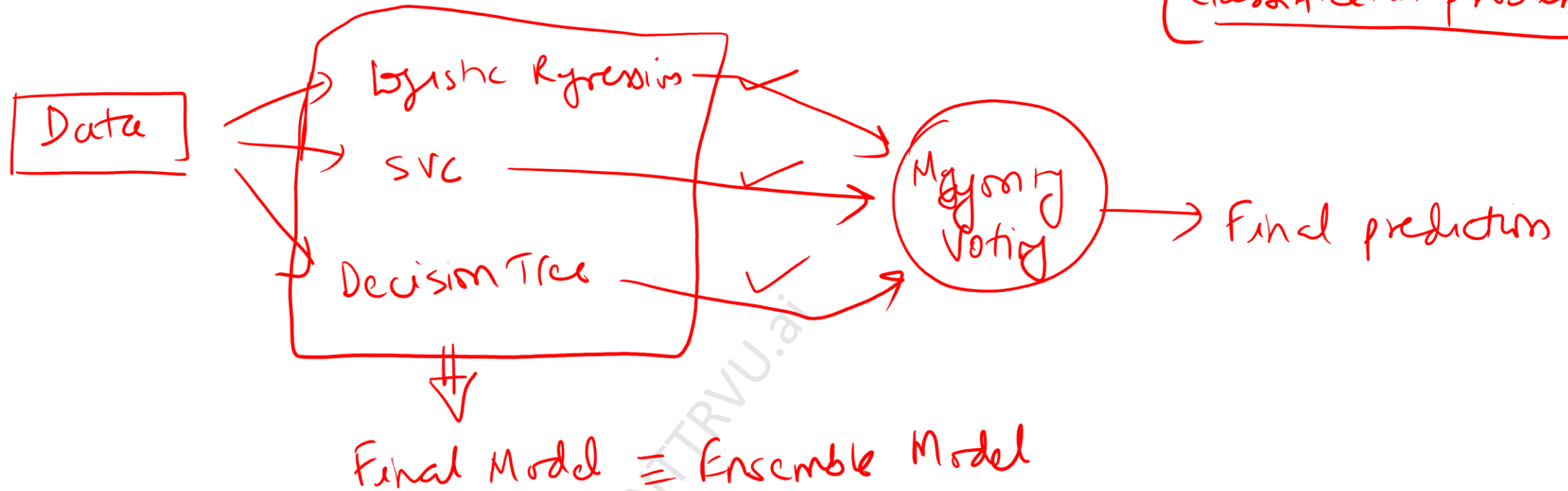
✓ Ensemble Models

Random Forest is a group of
multiple trees

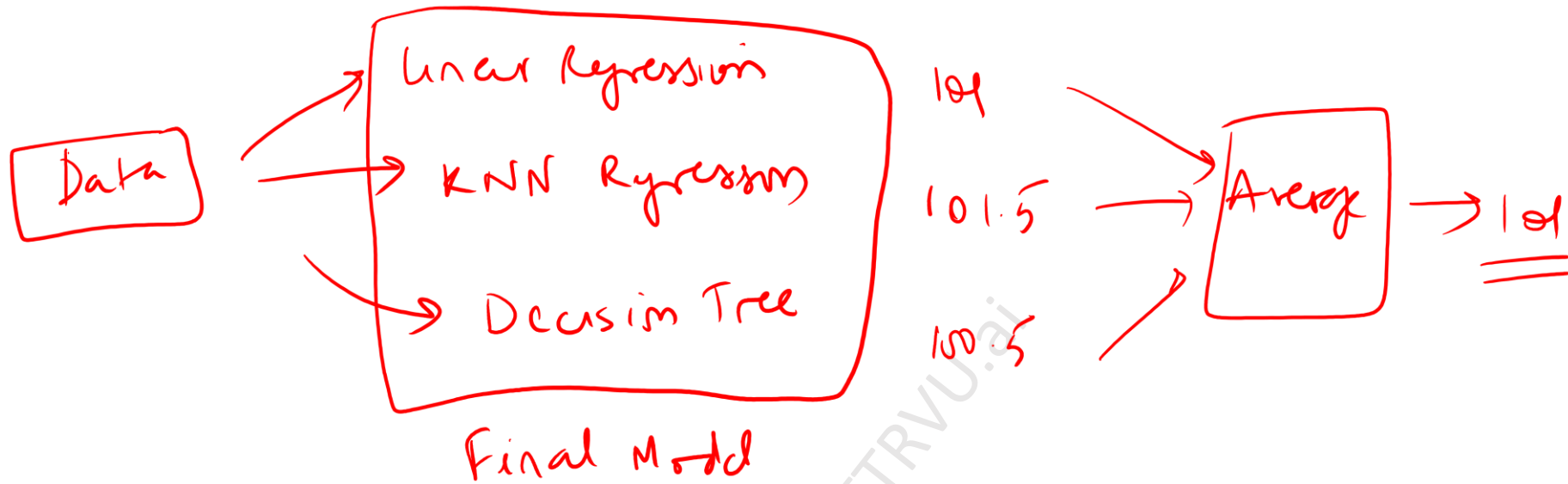
Random forest is an ensemble of multiple decision trees.

Ensemble Model is a group of various model

classification problems

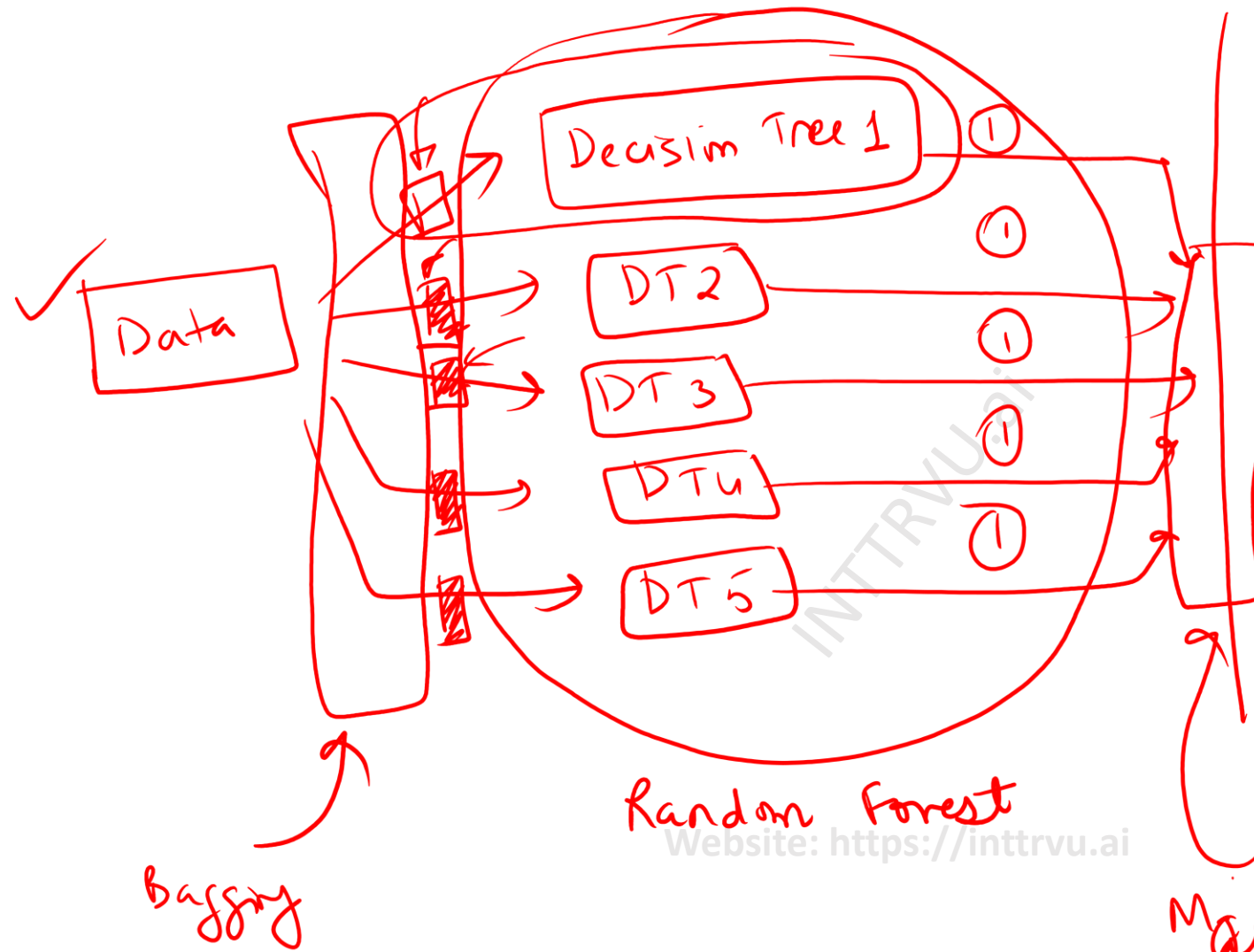


Regression Tasks Ensemble models



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Random forest is an ensemble of multiple decision trees



To make it more useful, we use different data

All the decision trees will be trained on different data

How to change the data for different decision trees?

the data is changed using a technique "BAGGING"

Data 1 to 100



We play a game; you have a random number generator from 1 to 100.

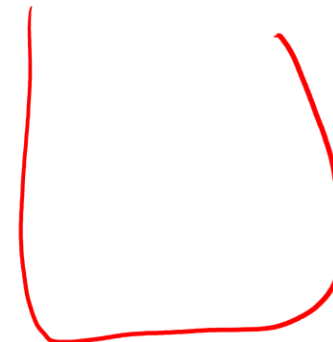
②, 90



BAG 1



BAG 2



BAG 3

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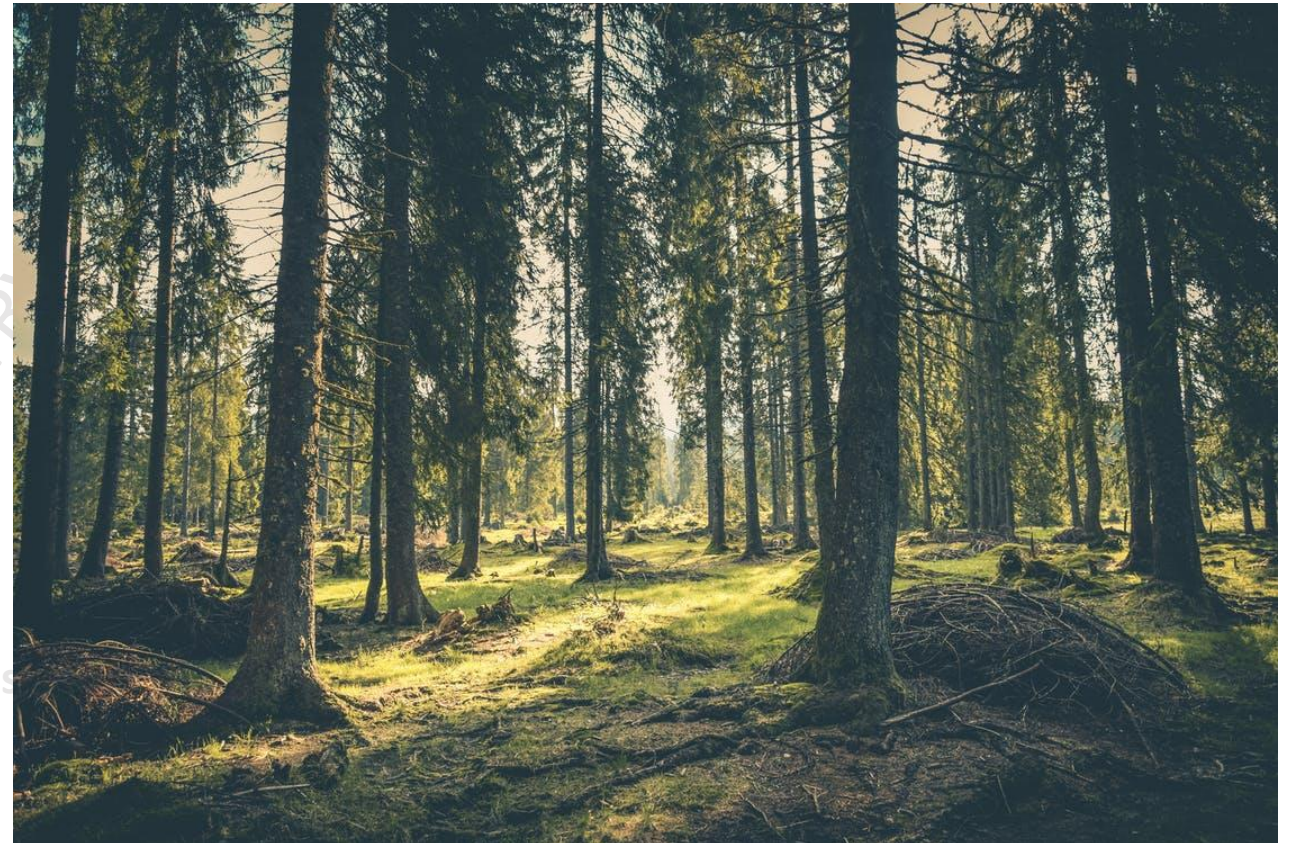
Random forest

- ① Random forests are not prone to overfitting

Random forest consists of multiple decision trees (forest)

Trees are created by randomly sampled dataset (rows) and features (columns)

- * * Trees are trained parallelly
This will reduce your training time.

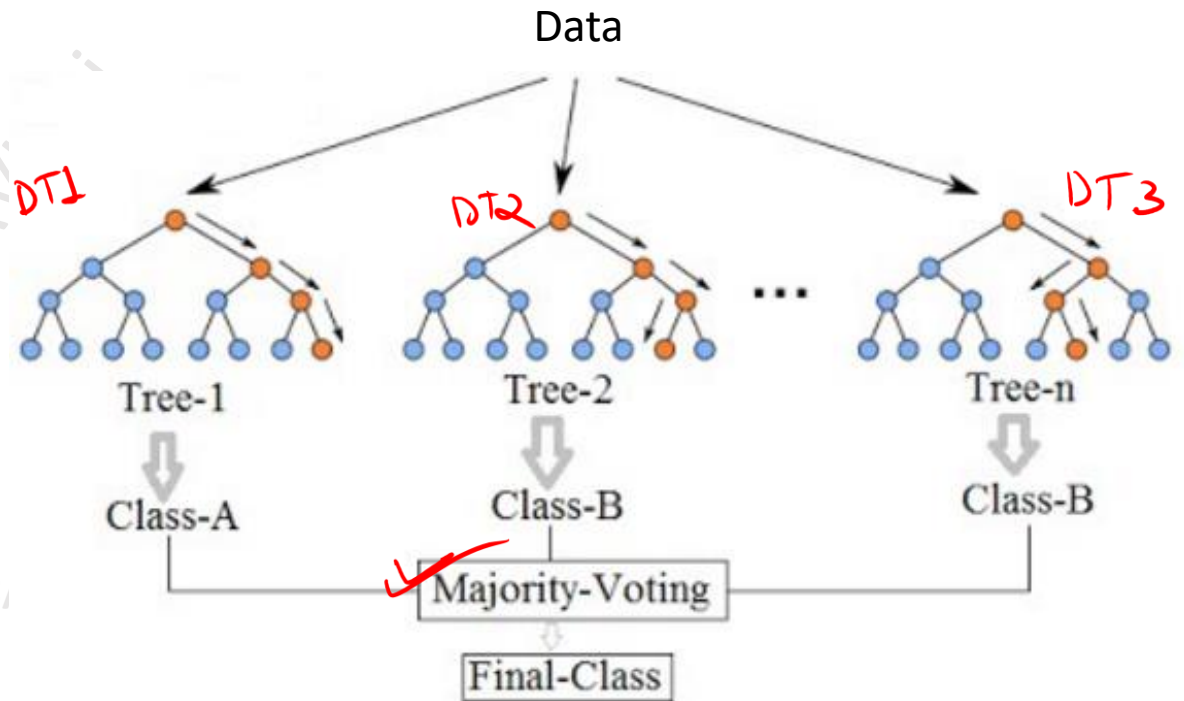


Random forest

Random forest is ensemble learning method

For classification, output of the random forest is the class selected by most trees.

For regression tasks, output is average of prediction of the individual trees



Bagging

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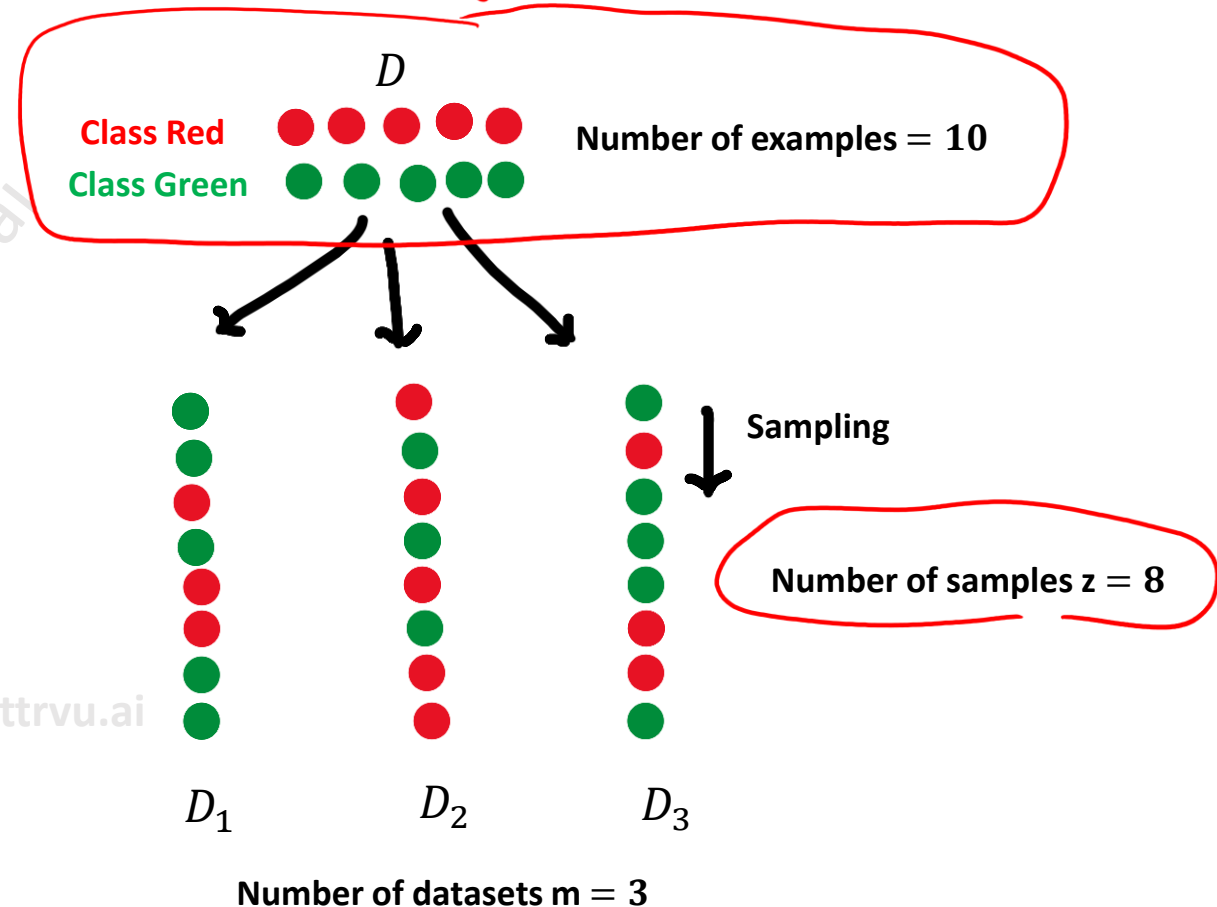
Bagging

Bagging for the row

Original

Bagging is technique of sampling training dataset D (consisting of n examples) to create m number of datasets D_i ($i = 1$ to m) each consisting of z examples

Sampling is done with replacement (sampled example is replaced back in dataset before extracting next sample)



Feature Bagging

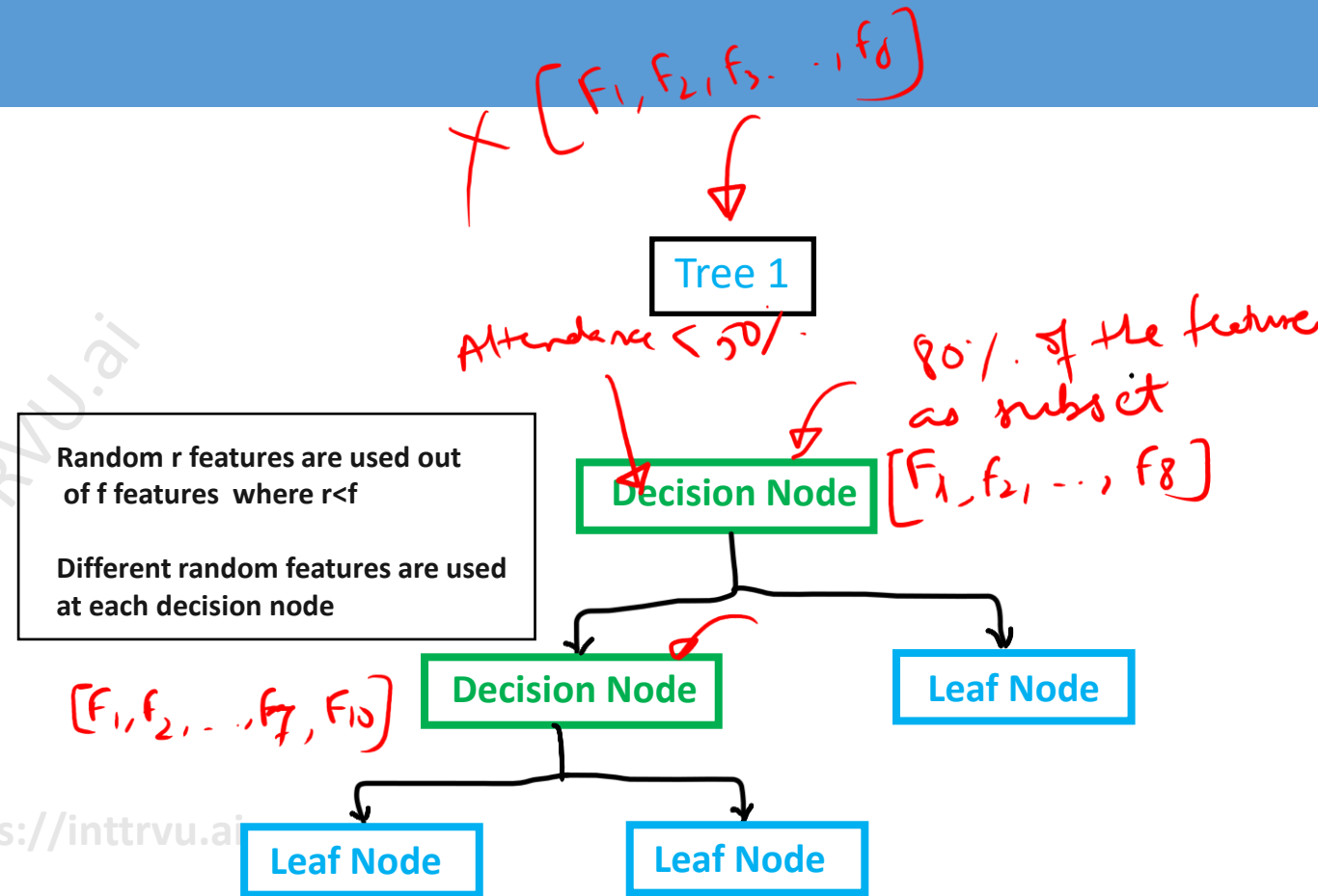
$F_1, F_2, F_3, F_4, \dots, F_{10}$

Feature bagging is a technique of selecting only subset of features while training each model (decision tree)

RF uses random subset of the features tree at each split in the learning process

If some features are very strong predictors of the output, these features will be used by many trees – resulting in correlated trees

Feature bagging results in low correlation between trees



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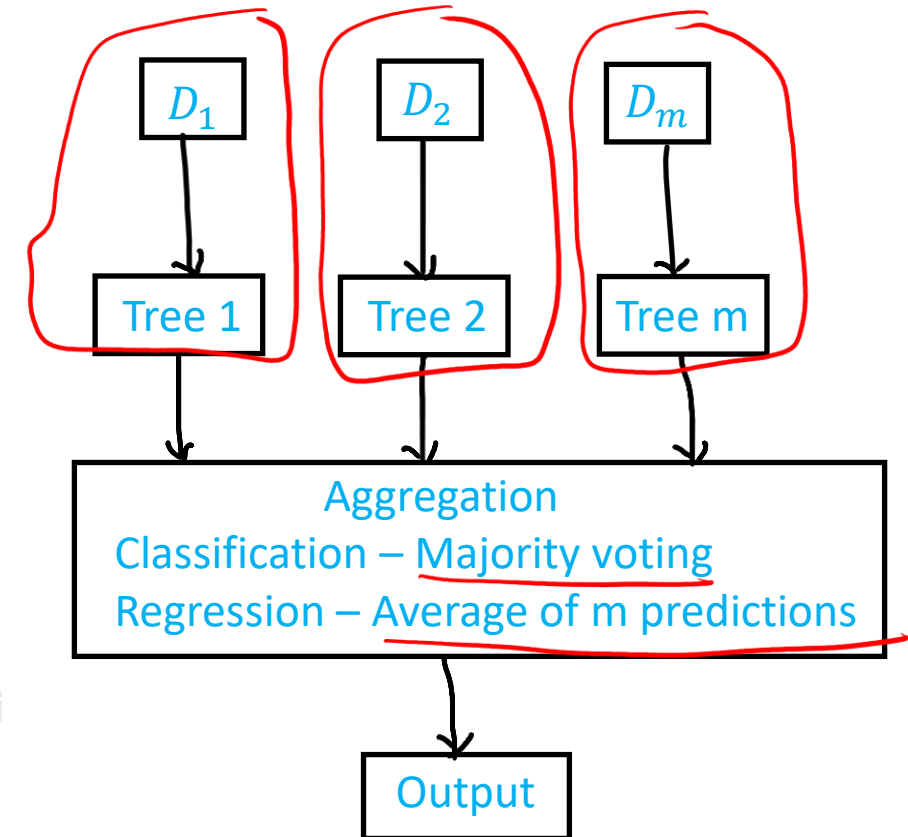
Random forest training

Bagging is also termed as bootstrap aggregating of the individual trees

Datasets obtained from bagging are used to train individual trees

For classification, output of the random forest is the class selected by most trees.

For regression tasks, output is average of prediction of the individual trees



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If-Else Blocks

Advantages of Random Forest

- Easy to interpret and explain to stakeholders
- Data scaling / normalization is not must
- Can learn linear as well as non-linear decision rules
- Bagging techniques used in RF results decreases variance without increasing bias of model
- RF are less prone to overfitting

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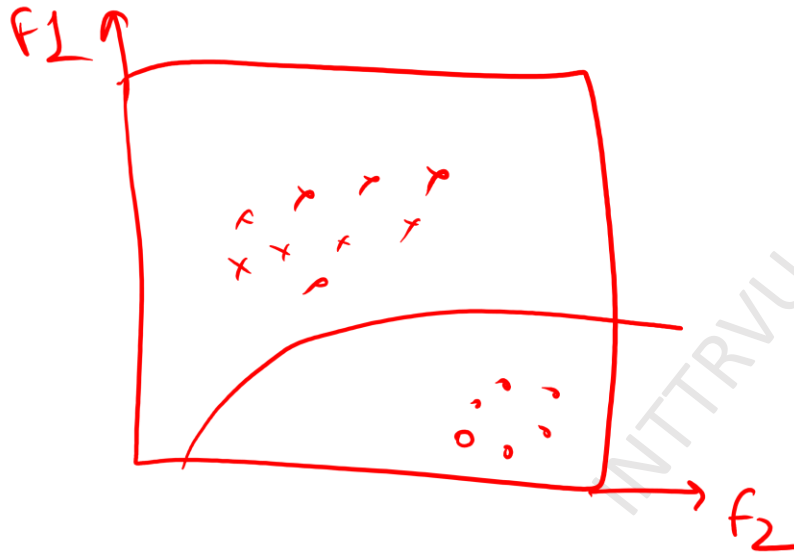
Explainable AI
Explainable models.

Feature Importance

Global Local

Global Feature Importance

Overall entire decision boundary



$$F_1 = \underline{\underline{51\%}}$$

$$F_2 = 10\%$$

$$F_8 = \underline{0.1\%}$$

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Local Feature Importance

Also important for model debugging

R5 → class 1 ✓

$$f_1 = +57\%$$

$$f_2 = -5\%$$

data drifting inspection
explaining your answers

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Feature Importance

Gini Importance / Mean Decrease in Impurity (MDI):

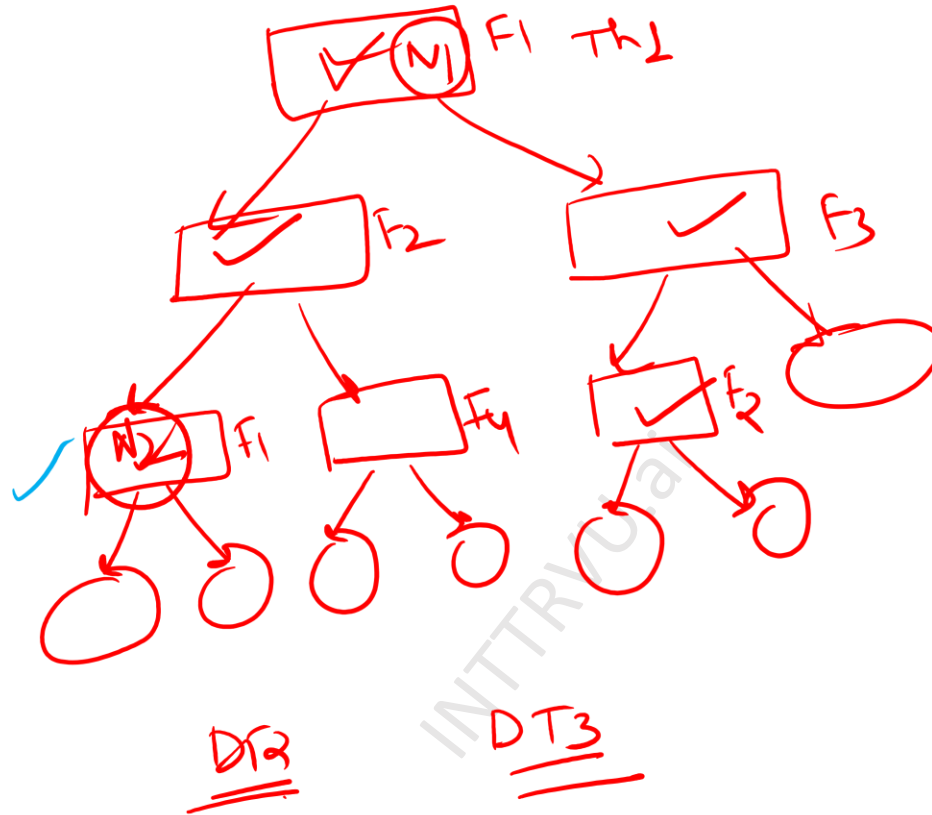
- It is average decreases in impurity for every feature computed across the trees
- Feature which decreases impurity more is of higher importance
- The impurity decrease for each feature is weighted by the number of data points in the corresponding node.

In random forest, say you have 100 decision trees

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DT1

$F_1 \Rightarrow \Delta G_1, \Delta G_2$
 $F_2 \Rightarrow \Delta G_3, \Delta G_4$
 $F_3 \Rightarrow \Delta G_5$
 $F_4 \Rightarrow \Delta G_6$



$F_1 =$ decrease in gini impurity

G before splitting

G_L, G_R

$\Delta G = G$ before splitting -

$$\left(\frac{N_L}{N_L + N_R} G_L + \frac{N_R}{N_L + N_R} G_R \right)$$



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DT1

$$F_1 \Rightarrow \Delta G_1, \Delta G_2 = \frac{N_1 \Delta G_1 + N_2 \Delta G_2}{N_1 + N_2} = \Delta G_1'$$

$$F_2 \Rightarrow \Delta G_3, \Delta G_4 = \frac{N_3 \Delta G_3 + N_4 \Delta G_4}{N_3 + N_4} = \Delta G_2'$$

$$F_3 \Rightarrow \Delta G_5$$

$$F_4 \Rightarrow \Delta G_6$$

DT2

$$F_1 \Rightarrow \Delta G$$

$$F_2 \Rightarrow \Delta G$$

$$\vdots$$

F_1 = average of ΔG across all the trees.

F_2

F_3

F_4



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Importance of a feature $\propto$ Reduction in Gini Impurity it causes.

$F_1 \rightarrow$	250
$F_2 \rightarrow$	190
$\vdots$	
$F_{10} \rightarrow$	10

$$F_1 > F_2 > \dots > F_{10}$$

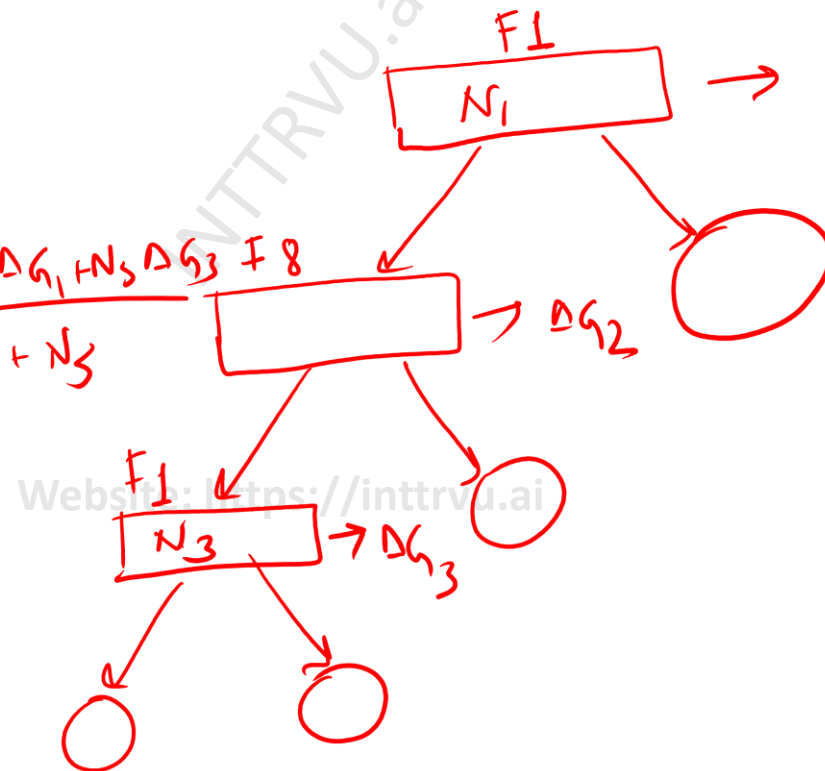
Importance $\longrightarrow$ decreases

DT1

$$F_1 \Rightarrow \Delta G_1,$$

$$\Delta G_3 = \frac{N_1 \Delta G_1 + N_5 \Delta G_3 + F_8}{N_1 + N_5}$$

$$\checkmark F_8 \Rightarrow \Delta G_2$$



G before splitting.

G_L, G_R

$\Delta G = G_{\text{before splitting}} -$

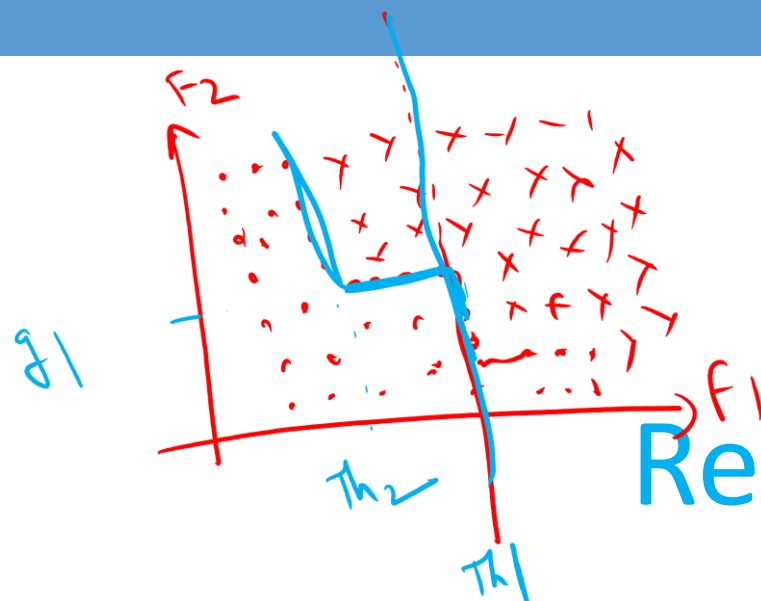
$$\left[\frac{N_L}{N_L + N_R} G_L + \frac{N_R}{N_L + N_R} G_R \right]$$

$DT_2, DT_3, \dots, DT_{100}$

$$\text{Feature for } F_1 \text{ Importance} = \frac{\Delta G_{DT_1}^{F_1} + \Delta G_{DT_2}^{F_1} + \dots + \Delta G_{DT_{100}}^{F_1}}{100}$$

$$\text{Feature of } f_2 \text{ Importance} = \frac{\Delta G_{DT_1}^{f_2} + \Delta G_{DT_2}^{f_2} + \dots + \Delta G_{DT_{100}}^{f_2}}{100}$$

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Regression using trees

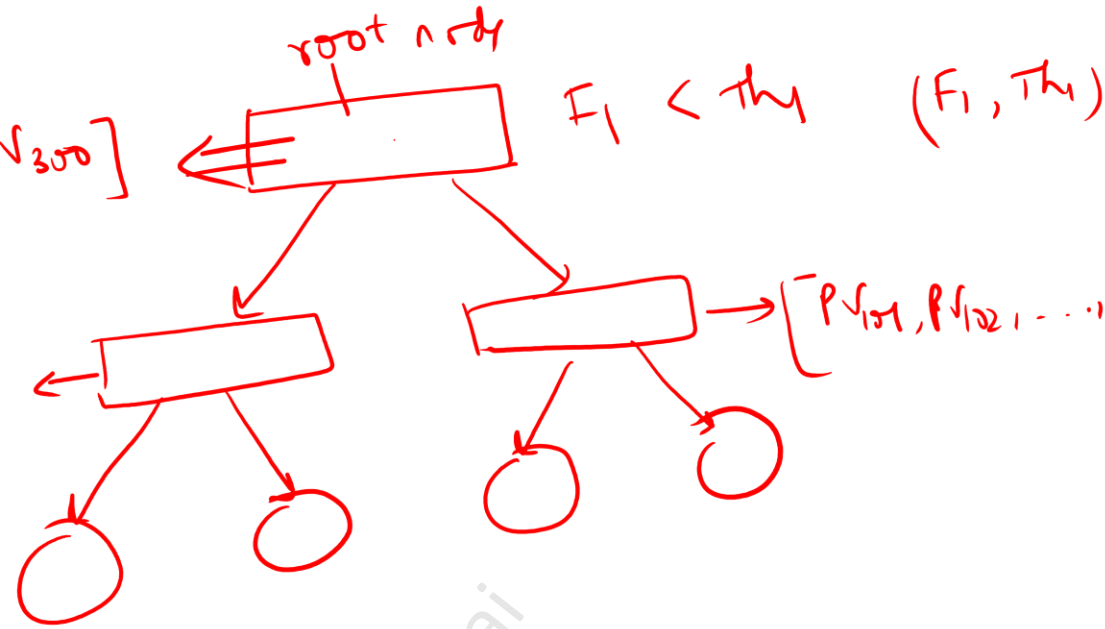
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Regression using trees

- Decision trees and Random Forest can also be used for regression models
- For regression problems, input data is partitioned based on reducing the variance of the value to be predicted
- To make a prediction for a new data point, it traverses the tree from the root to a leaf node based on the features of the data point.
- The prediction is typically the mean (or median) of the target values of the data points within that leaf.

array of numbers
 $\Rightarrow$ you can calculate variance

$[P_1, \dots, P_{100}]$



$$G = 1 - \sum_{i=1}^C p_i^2$$

Classification

Choose the feature which maximizes ΔG

$$\Delta G = G_{\text{before splitting}} - \left[\frac{N_L}{N_L + N_R} G_L + \frac{N_R}{N_L + N_R} G_R \right]$$

Regression

Choose the feature which maximizes ΔV ; V = variance of label y

$$\Delta V = V_{\text{before splitting}} - \left[\frac{N_L}{N_L + N_R} V_L + \frac{N_R}{N_L + N_R} V_R \right]$$

$$[PV_1, PV_2, \dots, PV_{300}] \Rightarrow \mu = \frac{\sum_{i=1}^{300} PV_i}{300}$$

$$V = \frac{1}{300} \sum_{i=1}^{300} (PV_i - \mu)^2$$

$(F_1, Th_1) \rightarrow \Delta V_1$
 $(F_1, Th_2) \rightarrow \Delta V_2$
 $(F_2, Th_3) \rightarrow \Delta V_3$

Choose that combination for which ΔV is maximum

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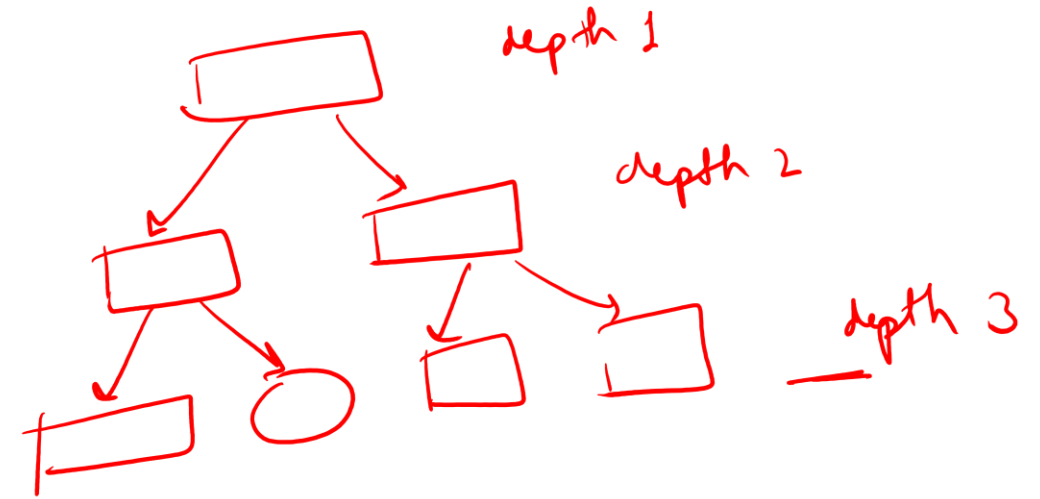
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Important Hyperparameters

① max-depth

max-depth = 3

To avoid overfitting



All data points belong to class 1

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② min-samples-split

min-samples-split = 10

data points = 5

No further splitting

2 values

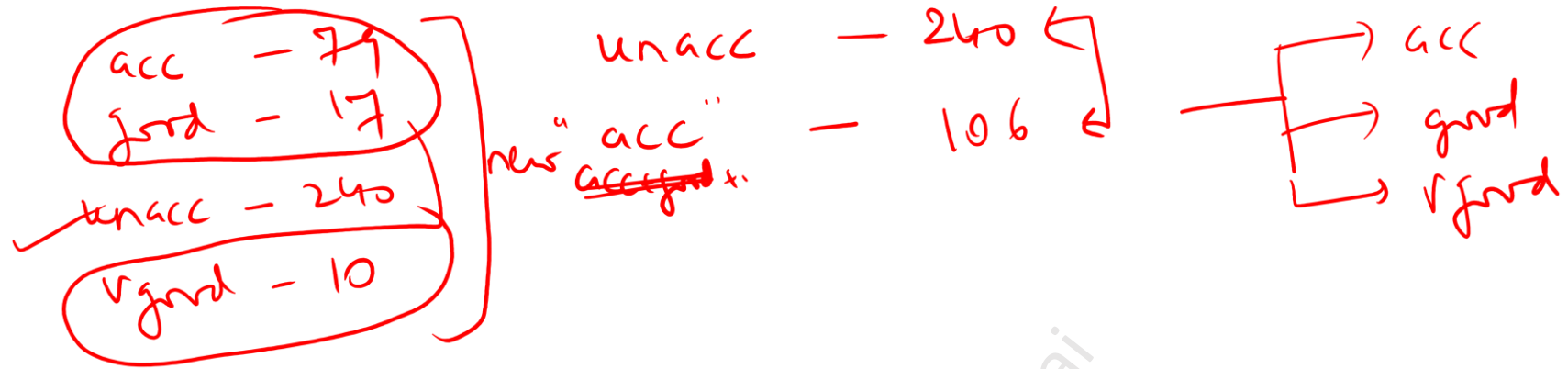
data points = 15

minimum no. of data points needed to split a node.

- ③ max-features $\Rightarrow$ used for feature bagging
- ④ max-samples $\Rightarrow$ used for row bagging

$$\frac{0.8}{0.8}$$

Multi-level modelling



Convert this multiclass problem to multiple binary problems

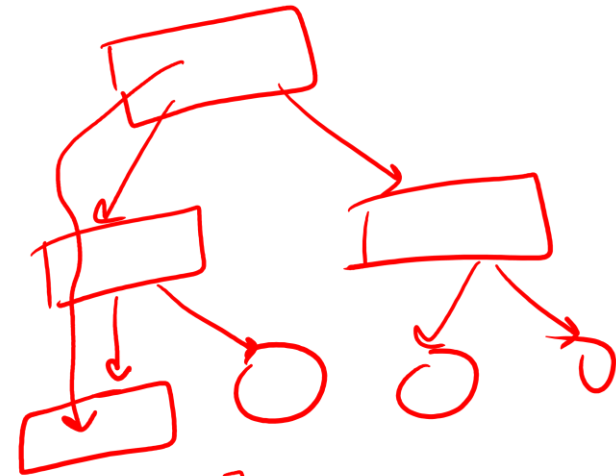
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⑤ n-estimators $\Rightarrow$ how many decision trees should be made in random forest

100

⑥ bootstrap : True or False
True : (default) $\Rightarrow$ with replacement
False $\Rightarrow$ w/o replacement

Questions?



$[P_1, P_2, \dots, P_N]$

final prediction = avg of all price values in the leaf node

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